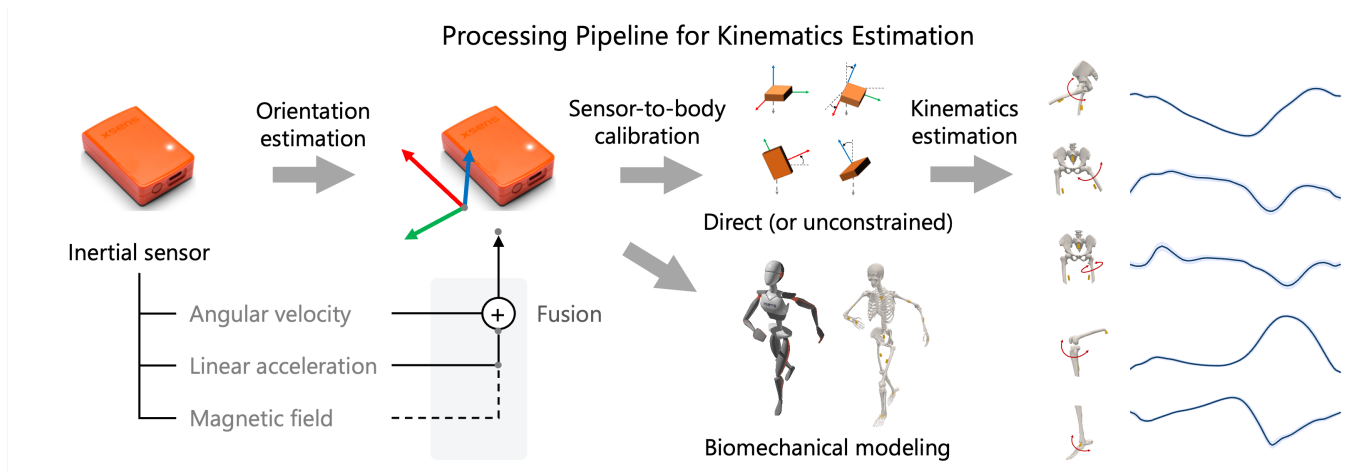




# Inertial Motion Tracking Matches Marker-Based Tracking Accuracy: Rethinking Modeling Approaches Towards Future Progress

preprint [link](#) data [IMOVE23](#) private data [In Review](#)

**IMULab** is a benchmark of state-estimation filters for tracking and studying human movements using inertial measurement units (IMUs). *A graphical user interface associated with this platform for no-code users will be released soon.*

This platform was initially created and validated for motion tracking of the lower body, but it can also be applied to the upper- or full-body motion tracking with some modifications.

## News

- ☒ IMU data from Experiments 1, 2, and 3 are now available on the [I-MOVE-23](#) page.
- ☐ The graphical user interface for converting IMU data to lower-limb kinematics will be released soon.

## Structure

See [imu\\_benchmark](#) for the implementation associated with Experiments 1, 2, and 3 presented in the paper; [imu\\_biodynamics](#) for Experiment 4.

## Installation

The code is tested with [Python 3.10](#) on [Ubuntu 24.04](#) and [MacOS](#). We recommend creating environments with [conda](#) to run the code.

```
conda create --name imu_benchmark python==3.10
conda activate imu_benchmark
```

See [AHRS](#), [VQF](#), and [RIANN](#) to install packages for state-estimation filters.

See [Scripting in Python](#) to install the `opensim` package for use of biomechanical modeling.

Run `python -m pip install -r requirements.txt` for other dependencies.

## Data

The [I-MOVE-23](#) dataset (Experiments 1, 2, and 3; see [imu\\_benchmark](#)) is now available for public use. This contains the data of Experiments 1, 2, and 3 mentioned in the paper. You will need to send a request with your institutional email account to download.

The [complete dataset](#), including the validation study with biplane fluoroscopy (Experiment 4; see [imu\\_biodynamics](#)), is shared privately with the reviewers, and will be made available upon publication of this manuscript.

See [README.md](#) in the [imu\\_benchmark](#) or [imu\\_biodynamics](#) folder for how to store the data.

## Run the Code

See [README.md](#) in the [imu\\_benchmark](#) or [imu\\_biodynamics](#) folder for how to run the code.

## Acknowledgements

We thank the authors of all the state-estimation filters included in this platform, for their contributions and for making their work open-sourced.

## Citation

If you use any part of the data or code, please cite [our paper](#).

```
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  author  = {Phan, Vu and Li, Zhixiong and Meinders, Evy and Gale, Tom and  
Anderst, William and Ng-Thow-Hing, Julian and Khandan, Aminreza and  
Halilaj, Eni},  
  journal = {In Submission},  
  year    = {2025}  
}
```

If you find this repository helpful, do not hesitate to give it a .

## Contact

Please contact Vu Phan ([vuphan@andrew.cmu.edu](mailto:vuphan@andrew.cmu.edu)) regarding any question and feedback.